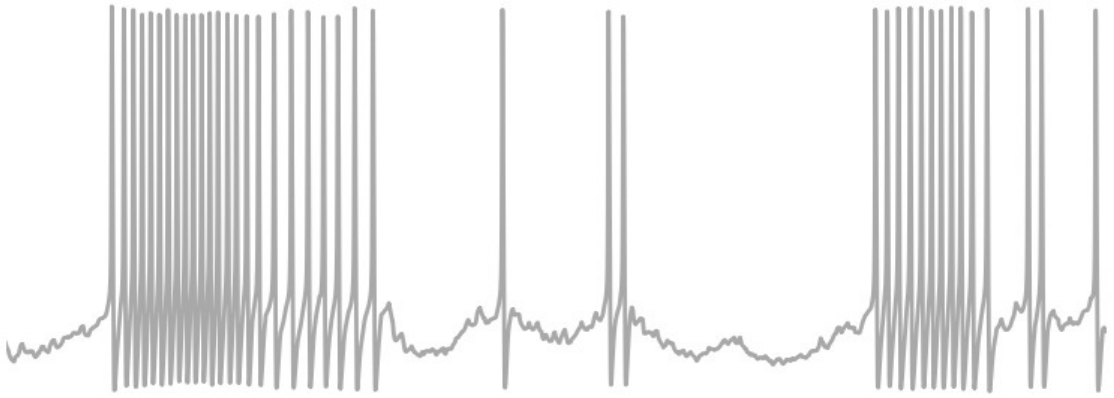
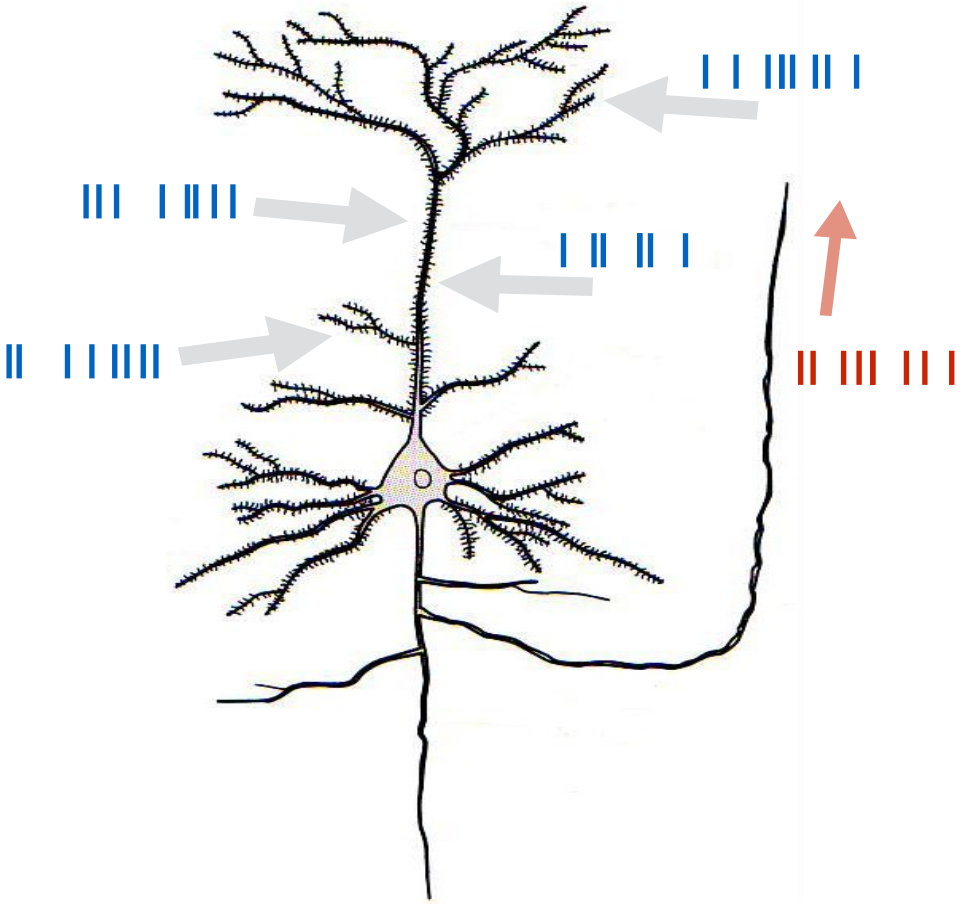


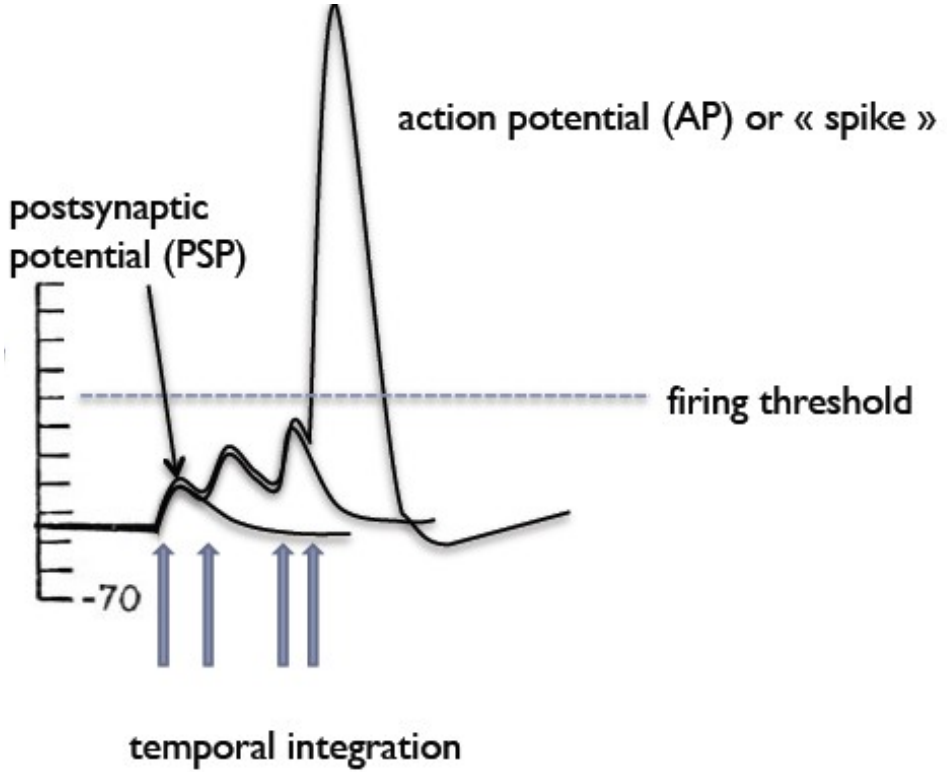
Single neuron simulations

Marcel Stimberg
ISIR (Sorbonne Université)

marcel.stimberg@sorbonne-universite.fr

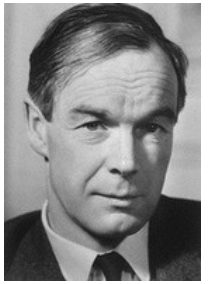
Neurons are basic units of computation



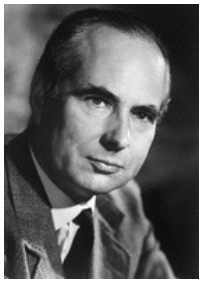


Neuron models

- **Hodgkin Huxley model** : description of ion channel dynamics (Hodgkin & Huxley, 1952)
- **integrate-and-fire model** : description of input integration membrane potential dynamics (LaPicque, 1907)



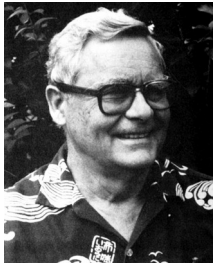
Hogdkin



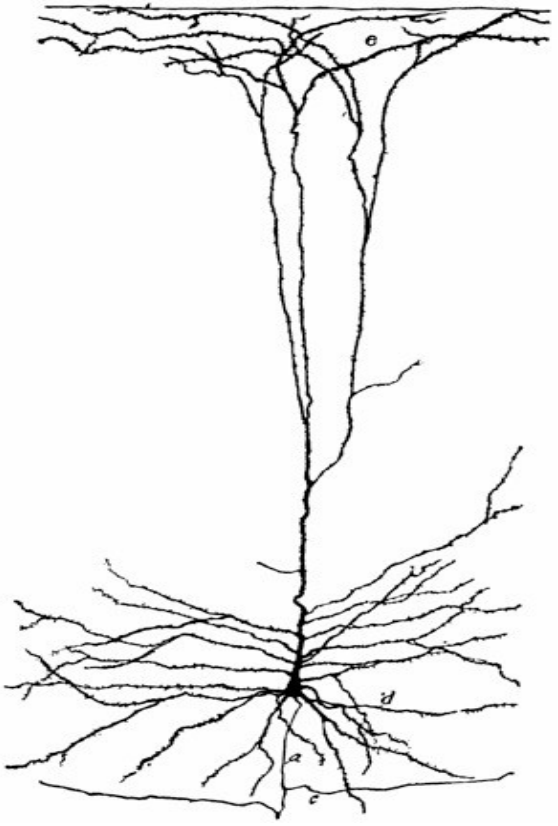
Huxley



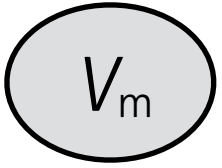
- **rate model** : description of the mean firing rate dynamics
- **cable theory** : description of input propagation along the dendrites (Rall, 1962)



Simplified models: point neuron

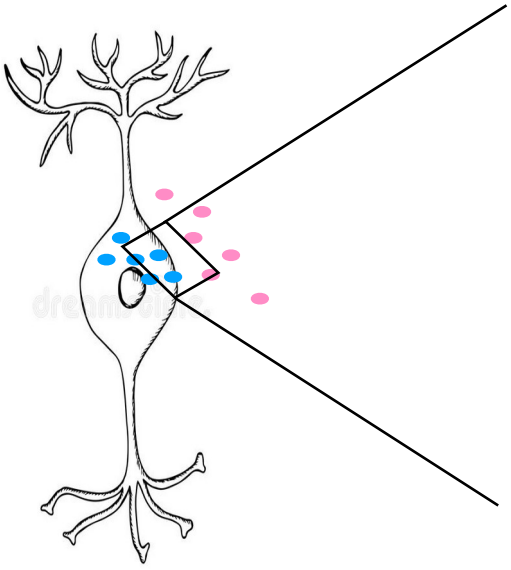




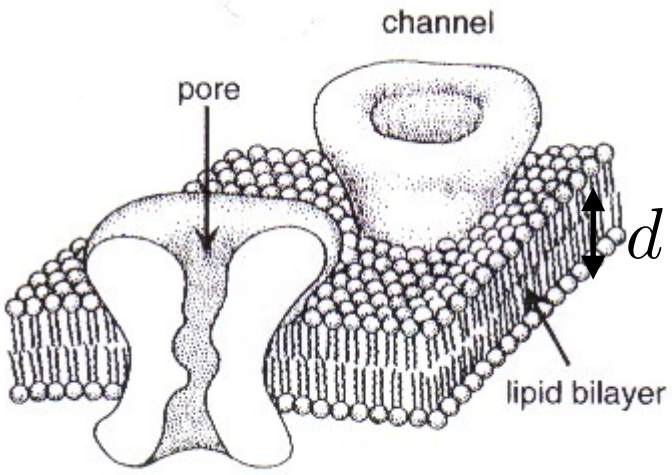


The neuronal cell membrane

The cell membrane is a lipid bilayer with protein inclusions



extracellular charges
(mostly Na^+, Cl^-)



$d \simeq 3$ nm

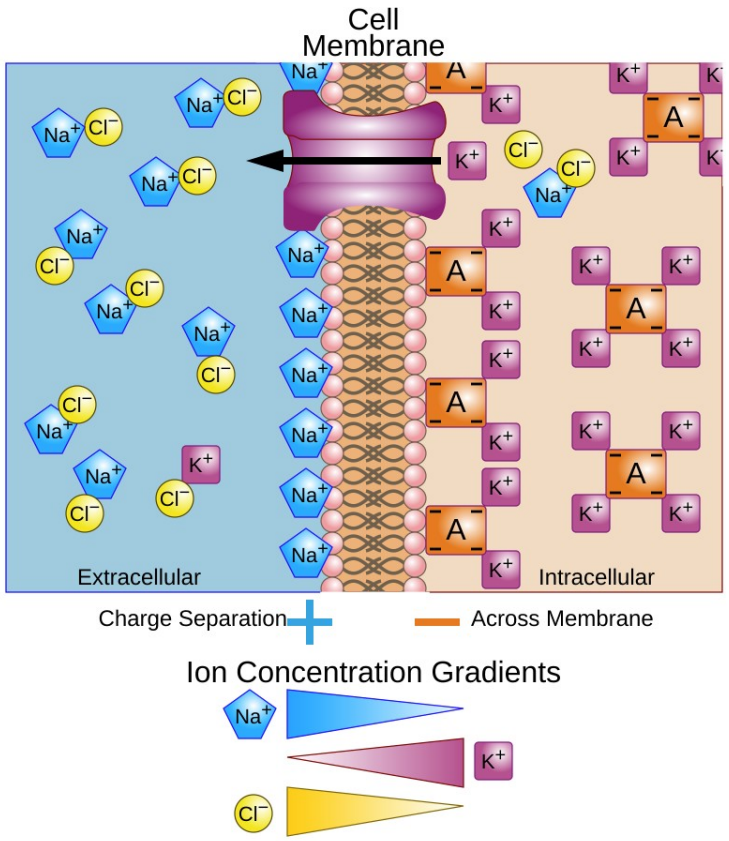
intracellular charges
(mostly K^+)

$$[K^+]_{\text{out}} \simeq 4\,\text{mM}$$
$$[Na^+]_{\text{out}} \simeq 144\,\text{mM}$$

$$[K^+]_{\text{in}} \simeq 160\,\text{mM}$$
$$[Na^+]_{\text{in}} \simeq 12\,\text{mM}$$

The neuronal cell membrane

Each ion type has its **reversal potential**, where **concentration gradient** and **electrical gradient** cancel each other out

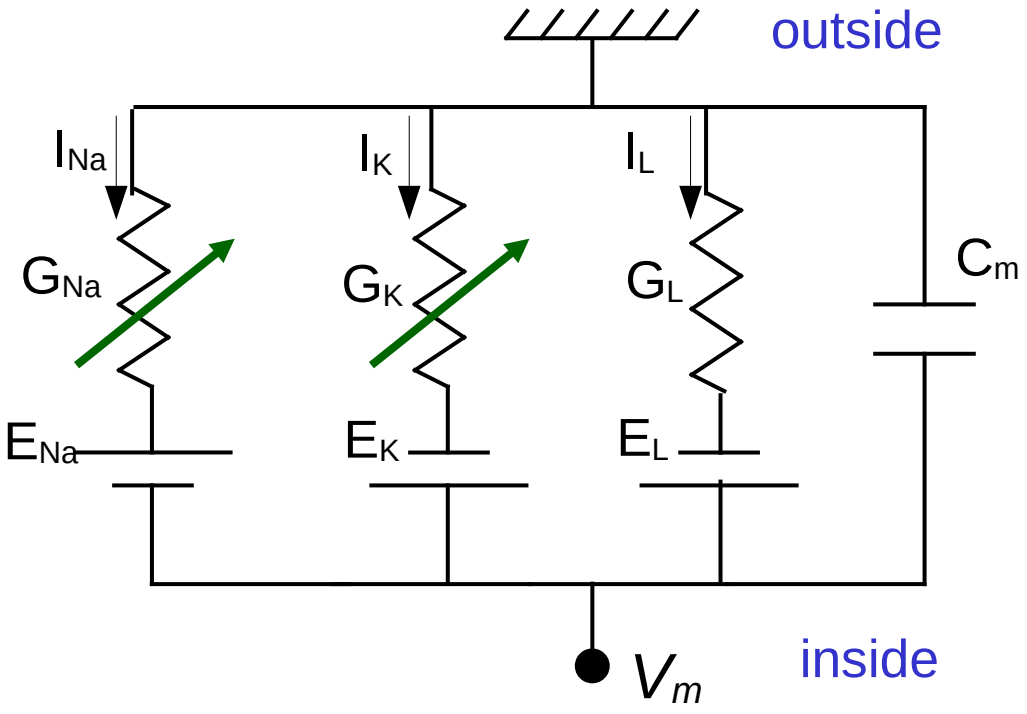


Reversal potentials (approx.)

Sodium (**Na**): +50mV
Potassium (**K**): -90mV
Chloride (**Cl**): -80mV

The neuronal cell membrane

Electrical equivalent circuit



Kirchhoff's law:

$IC_m + I_{Na} + I_K + I_L = 0$

Definition of capacitance:

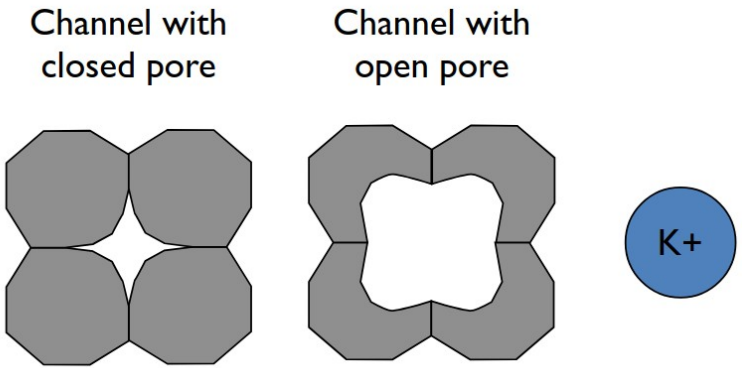
$IC_m = C^{dV}/_{dt}$

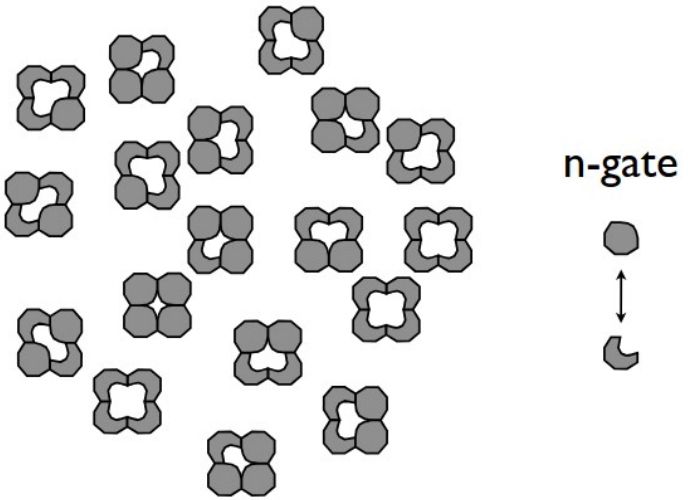
Ohm's law:

$I_x = g_x(V_m - E_x)$

Driving force

HH model: the potassium channel



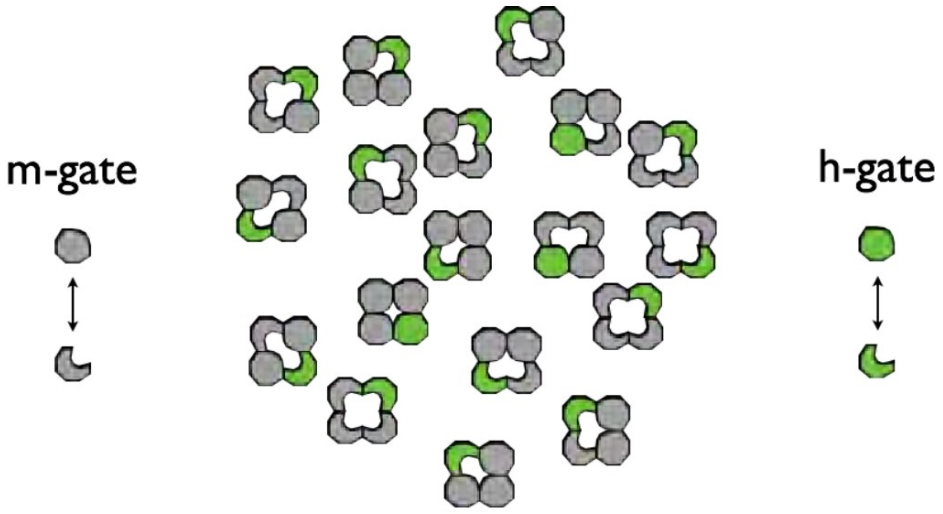


4 similar "subunits" need to be open

Probability of a subunit being open = n
Probability of channel being open = n^4

$$I_K = \bar{g}_K n^4 (V_m - E_K)$$

HH model: the sodium channel

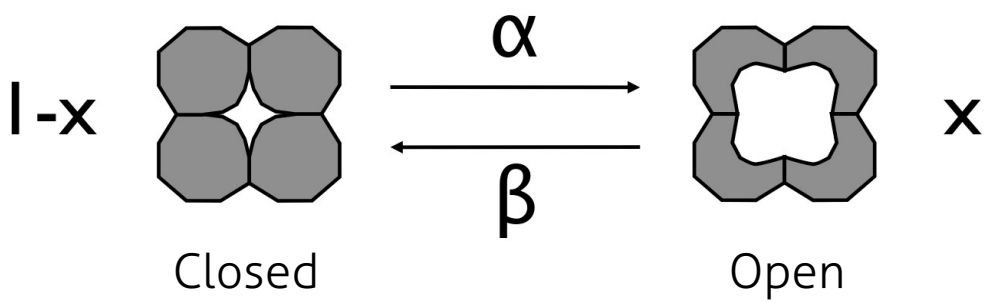


Two different types of subunits

$$I_{Na} = \bar{g}_{Na} m^3 h (V_m - E_{Na})$$

HH model: channel kinetics

x = probability that a channel is open
= fraction of channels that is open



$$dx/dt = \alpha (1 - x) - \beta x$$

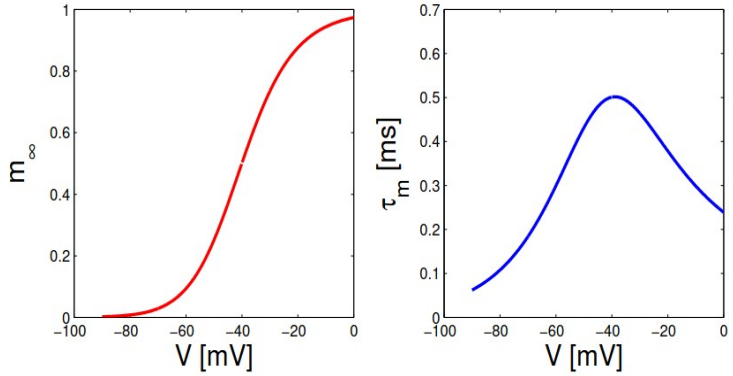
Re-arranging terms:

$$\underbrace{1/(\alpha + \beta)}_{\tau_x}\, dx/dt = \underbrace{\alpha/(\alpha + \beta)}_{x_\infty} - x$$

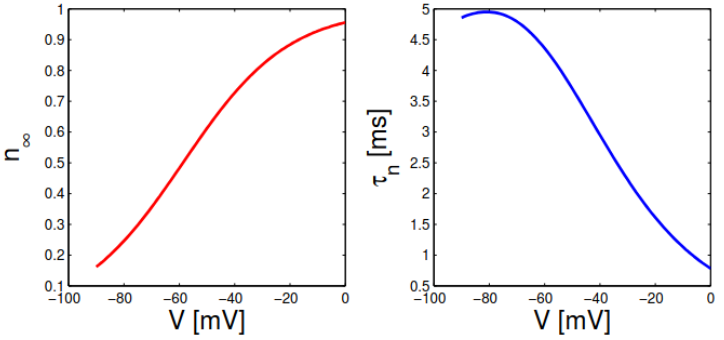
$$\tau_x\, dx/dt = x_\infty - x$$

HH model: channel kinetics

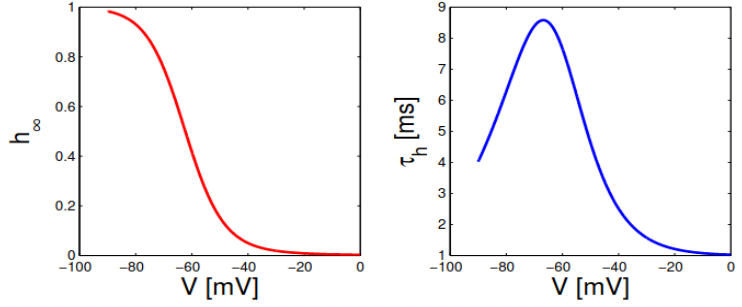
m-activation **increases rapidly** with increasing membrane voltage.



n-activation **increases slowly** with increasing membrane voltage.



h-activation **decreases slowly** with increasing membrane voltage



The full Hodgkin-Huxley model equations

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = g_L(E_L - V) + \bar{g}_{\mathrm{Na}}m(t)^3h(t)(E_{\mathrm{Na}} - V) + \bar{g}_{\mathrm{K}}n(t)^4(E_{\mathrm{K}} - V) + I_{\mathrm{stim}}$$

$$\tau_n\frac{\mathrm{d}n}{\mathrm{d}t} = n_\infty - n \qquad \tau_m\frac{\mathrm{d}m}{\mathrm{d}t} = m_\infty - m \qquad \tau_h\frac{\mathrm{d}h}{\mathrm{d}t} = h_\infty - h$$

$$\tau_n = \frac{1}{\alpha_n + \beta_n} \qquad \tau_m = \frac{1}{\alpha_m + \beta_m} \qquad \tau_h = \frac{1}{\alpha_h + \beta_h}$$

$$n_\infty = \frac{\alpha_n}{\alpha_n + \beta_n} \qquad m_\infty = \frac{\alpha_m}{\alpha_m + \beta_m} \qquad h_\infty = \frac{\alpha_h}{\alpha_h + \beta_h}$$

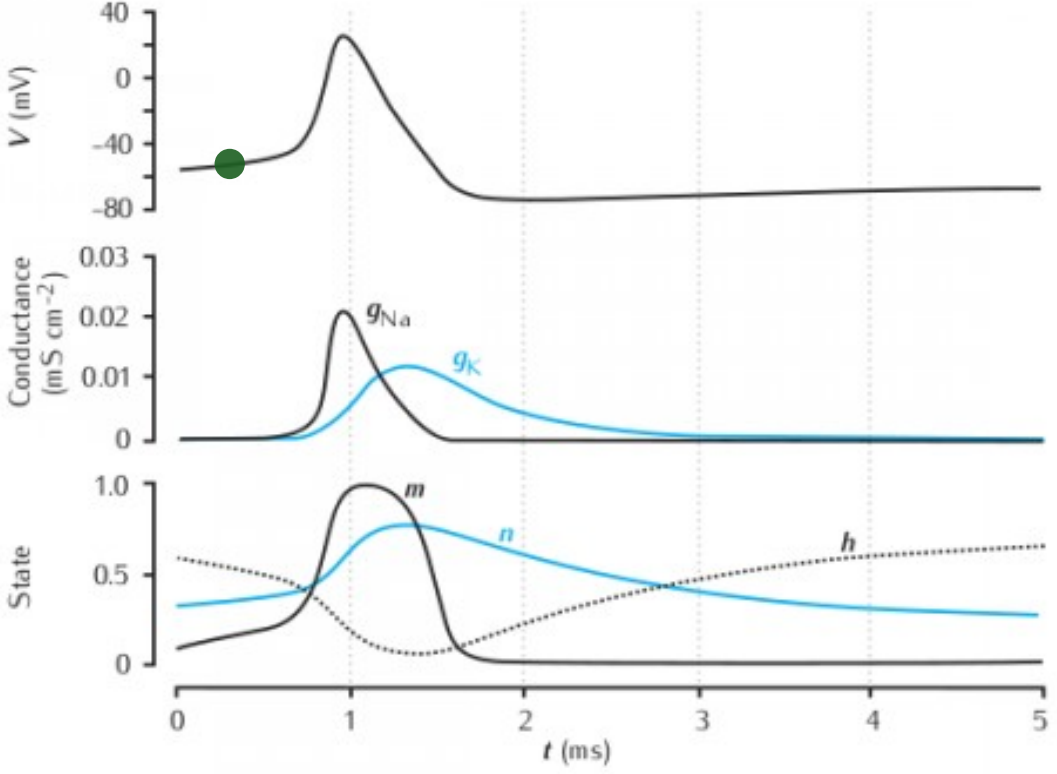
Transition rates (parameter values determined for squid giant axon):

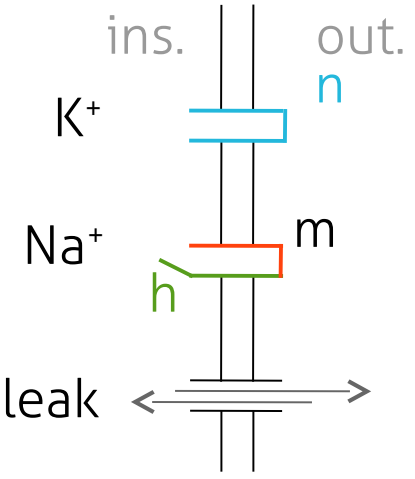
$$\alpha_n(V) = \frac{(0.1 - 0.01V)}{e^{1-0.1V} - 1} \qquad \alpha_m(V) = \frac{(2.5 - 0.1V)}{e^{2.5-0.1V} - 1} \qquad \alpha_h(V) = 0.07\,e^{-\frac{V}{20}}$$

$$\beta_n(V) = 0.125e^{-\frac{V}{80}} \qquad \beta_m(V) = 4e^{-\frac{V}{18}} \qquad \beta_h(V) = \frac{1}{e^{3-0.1V} + 1}$$

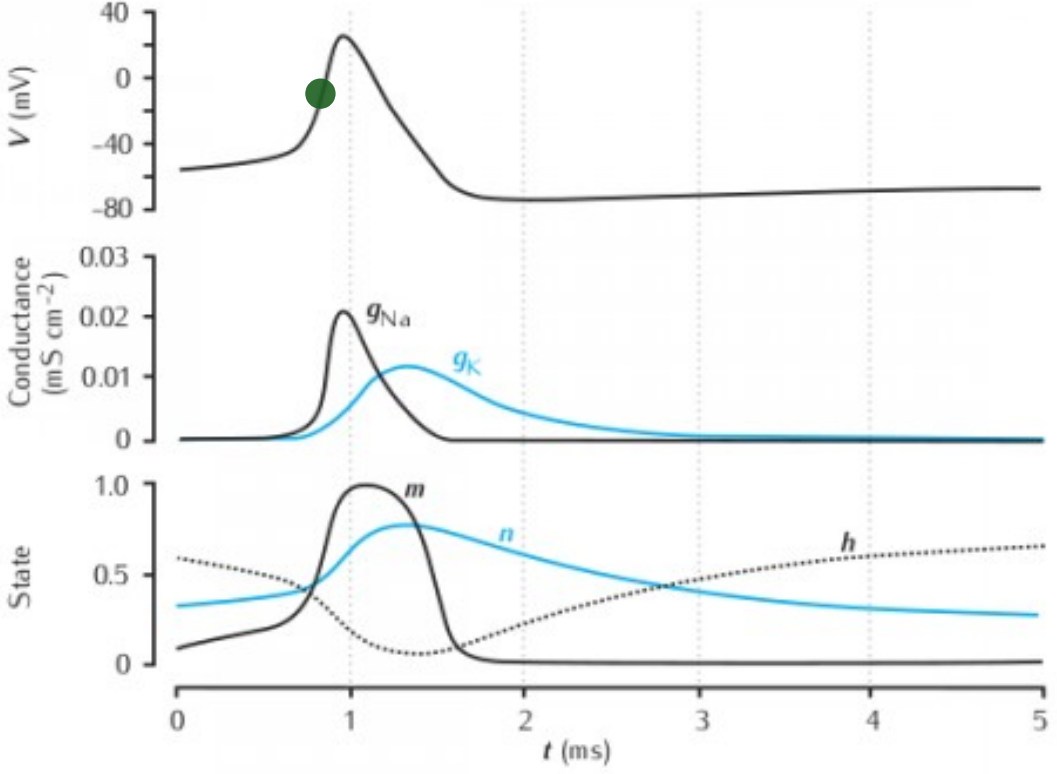
(**Rates** expressed as **/ms**, membrane potential **V in mV**)

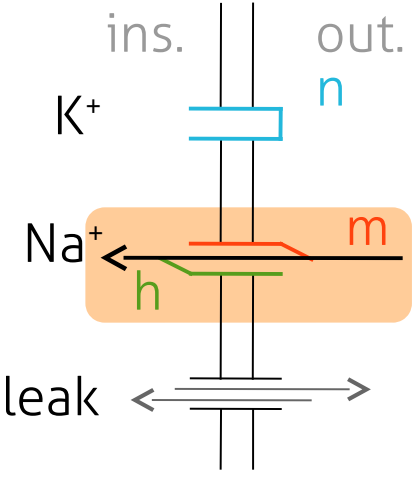
HH model: the action potential



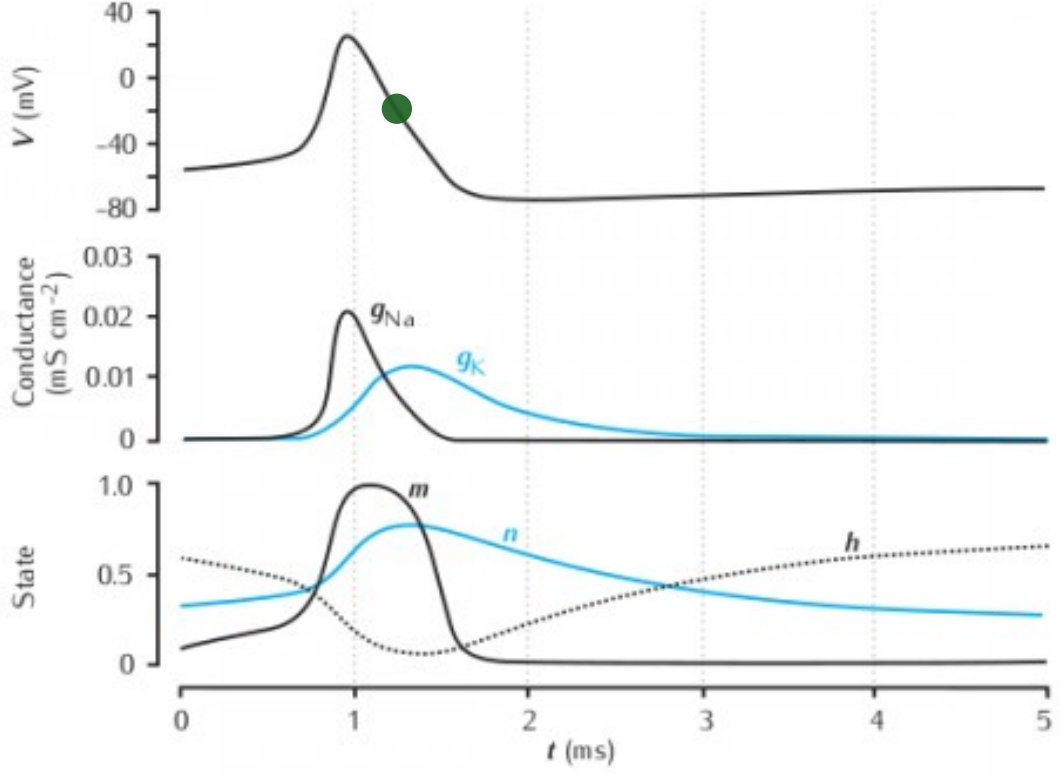


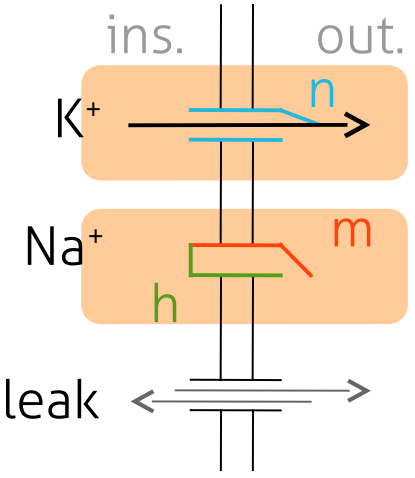
HH model: the action potential



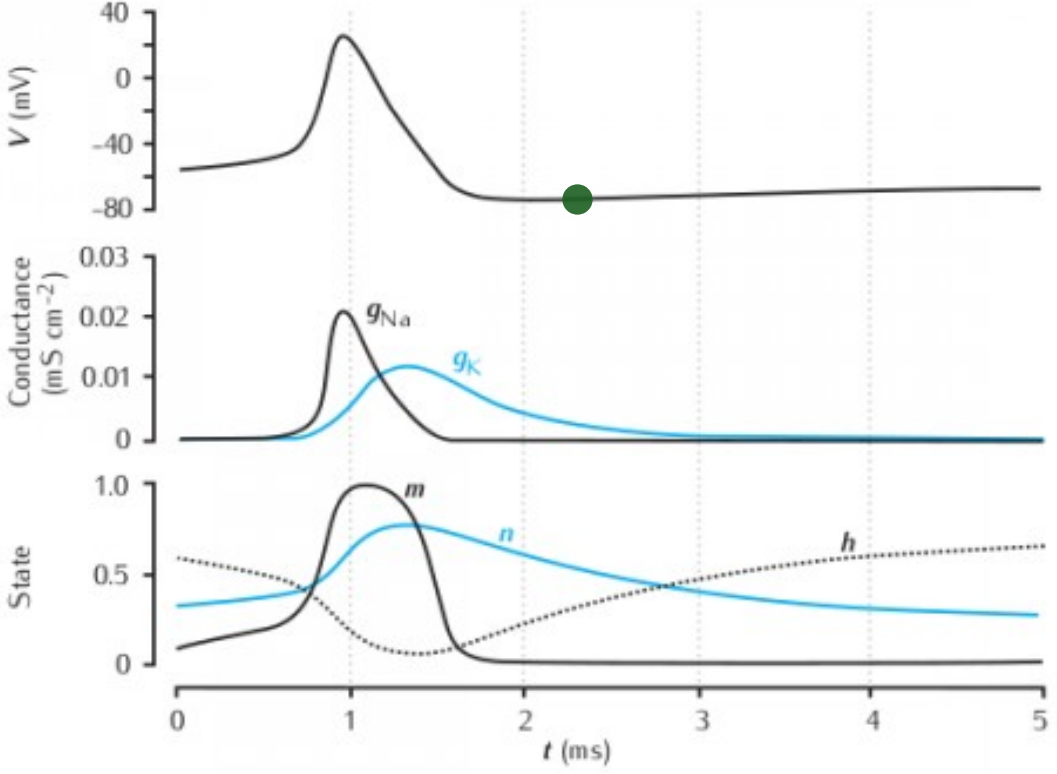


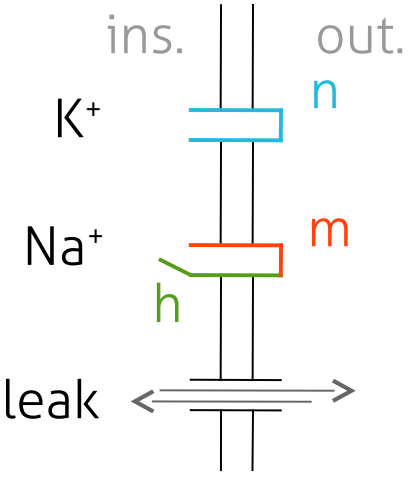
HH model: the action potential





HH model: the action potential





Variants of the Hodgkin-Huxley model

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = g_L(E_L - V) + \bar{g}_{\mathrm{Na}}m(t)^3h(t)(E_{\mathrm{Na}} - V) + \bar{g}_{\mathrm{K}}n(t)^4(E_{\mathrm{K}} - V) + I_{\mathrm{stim}}$$
$$+ \bar{g}_{\mathrm{KLT}}w(t)^4z(t)(E_K - V)$$

$$\tau_n\frac{\mathrm{d}n}{\mathrm{d}t} = n_\infty - n \qquad \tau_m\frac{\mathrm{d}m}{\mathrm{d}t} = m_\infty - m \qquad \tau_h\frac{\mathrm{d}h}{\mathrm{d}t} = h_\infty - h$$

$$\tau_n = \frac{1}{\alpha_n + \beta_n} \qquad \tau_m = \frac{1}{\alpha_m + \beta_m} \qquad \tau_h = \frac{1}{\alpha_h + \beta_h}$$

$$n_\infty = \frac{\alpha_n}{\alpha_n + \beta_n} \qquad m_\infty = \frac{\alpha_m}{\alpha_m + \beta_m} \qquad h_\infty = \frac{\alpha_h}{\alpha_h + \beta_h}$$

$$\tau_w\frac{\mathrm{d}w}{\mathrm{d}t} = w_\infty - w \qquad \tau_z\frac{\mathrm{d}z}{\mathrm{d}t} = z_\infty - z$$

$$\tau_w = \frac{1}{\alpha_w + \beta_w} \qquad \tau_z = \frac{1}{\alpha_z + \beta_z}$$

$$w_\infty = \frac{\alpha_w}{\alpha_w + \beta_w} \qquad z_\infty = \frac{\alpha_z}{\alpha_z + \beta_z}$$

Adding new channels, following the same formalism
Often different types of Potassium channels, or Calcium channels

Leaky integrate-and-fire (LIF) model

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = g_L(E_L - V) + \bar{g}_{\mathrm{Na}}m(t)^3h(t)(E_{\mathrm{Na}} - V) + \bar{g}_{\mathrm{K}}n(t)^4(E_{\mathrm{K}} - V) + I_{\mathrm{stim}}$$

$$\tau_n\frac{\mathrm{d}n}{\mathrm{d}t} = n_\infty - n \qquad \tau_m\frac{\mathrm{d}m}{\mathrm{d}t} = m_\infty - m \qquad \tau_h\frac{\mathrm{d}h}{\mathrm{d}t} = h_\infty - h$$

$$\tau_n = \frac{1}{\alpha_n + \beta_n} \qquad \tau_m = \frac{1}{\alpha_m + \beta_m} \qquad \tau_h = \frac{1}{\alpha_h + \beta_h}$$

$$n_\infty = \frac{\alpha_n}{\alpha_n + \beta_n} \qquad m_\infty = \frac{\alpha_m}{\alpha_m + \beta_m} \qquad h_\infty = \frac{\alpha_h}{\alpha_h + \beta_h}$$

Transition rates (parameter values determined for squid giant axon):

$$\alpha_n(V) = \frac{(0.1 - 0.01V)}{e^{1-0.1V} - 1} \qquad \alpha_m(V) = \frac{(2.5 - 0.1V)}{e^{2.5-0.1V} - 1} \qquad \alpha_h(V) = 0.07\,e^{-\frac{V}{20}}$$

$$\beta_n(V) = 0.125e^{-\frac{V}{80}} \qquad \beta_m(V) = 4e^{-\frac{V}{18}} \qquad \beta_h(V) = \frac{1}{e^{3-0.1V} + 1}$$

(**Rates** expressed as **/ms**, membrane potential **V in mV**)

Leaky integrate-and-fire (LIF) model

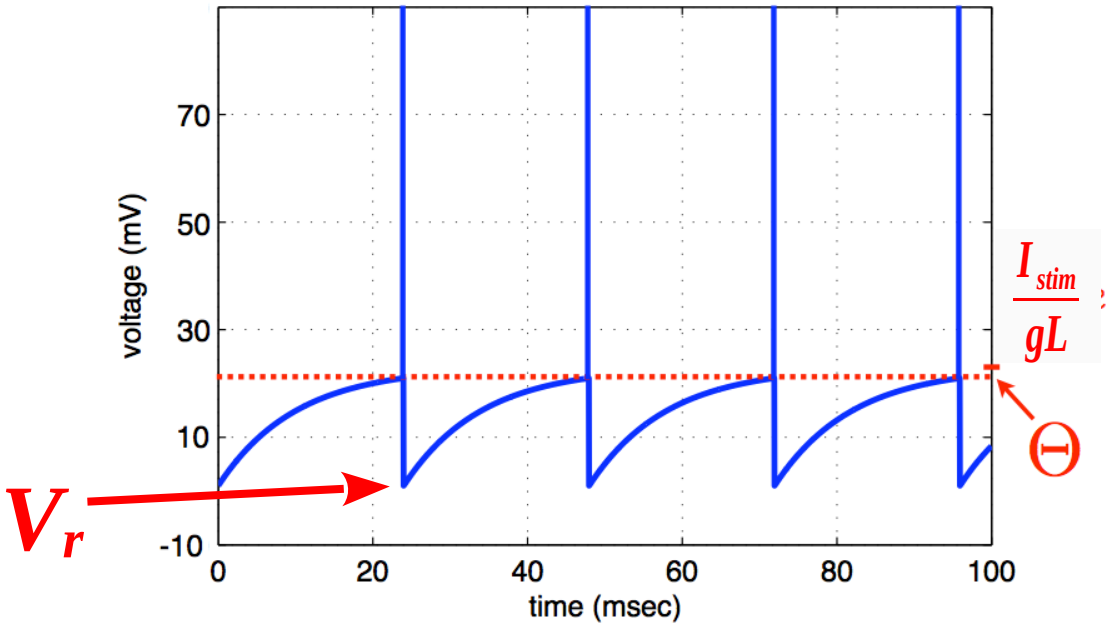
$$C\frac{\mathrm{d}V}{\mathrm{d}t} = g_L(E_L - V) + I_{\mathrm{stim}}$$

+

Threshold: emit a spike if *V* crosses Θ

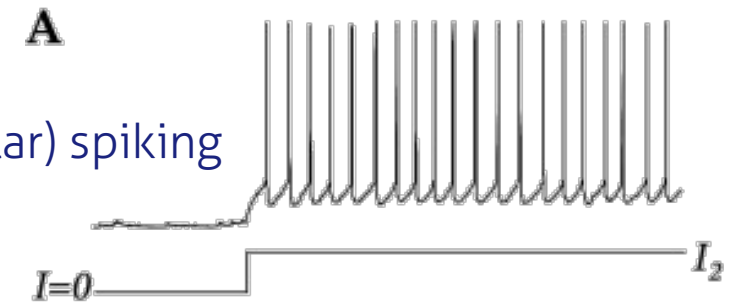
Reset: reset the membrane potential to V_r

(Refractoriness: do not allow spikes for *x* ms after a spike)

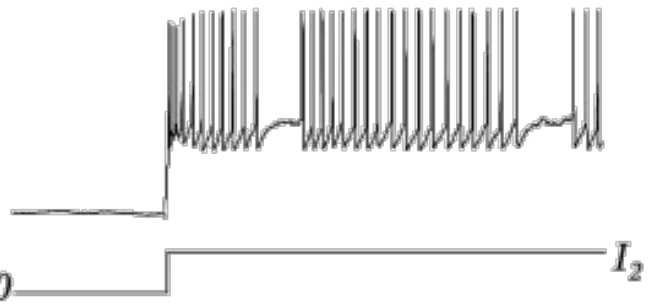


Limitations of the LIF model

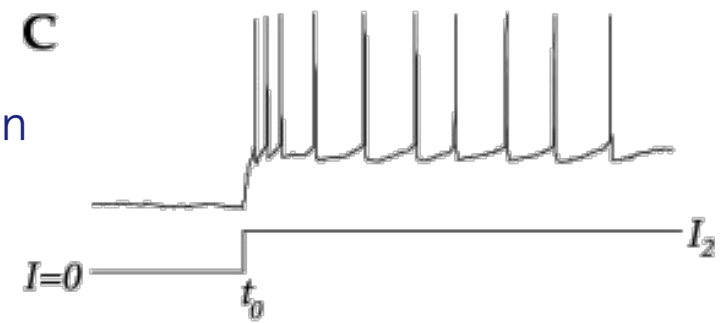
Activity is reset after each spike, no "memory" of previous input or activity
→ Cannot model bursts or adaptation as seen in real neurons



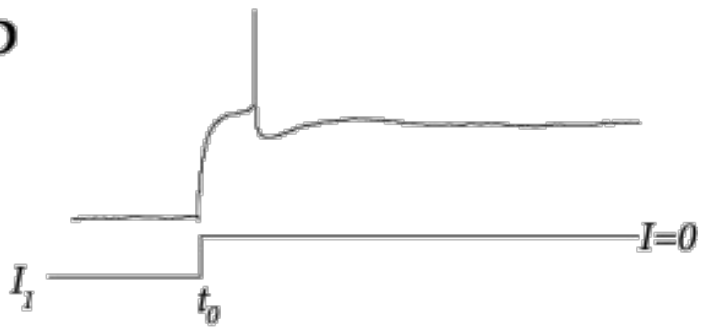
Fast (regular) spiking



"Stuttering"



Adaptation



Inhibitory Rebound

Neural Dynamics text book (Chapter 1.4)

Data courtesy of Henry Markram and Maria Toledo-Rodriguez

Extensions of the LIF model

LIF + adaptation current

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = \ldots + w$$

$$\tau_w\frac{\mathrm{d}w}{\mathrm{d}t} = a(E_L - V) - w$$

Threshold: emit a spike if V crosses Θ
Reset: reset the membrane potential to V_r
increase w by b

Second variable: slow adaptation current
a determines sub-threshold adaptation, b determines spike-triggered adapation

Examples: adaptive exponential integrate-and-fire model, Izhikevich model

LIF + adaptative threshold

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = g_L(E_L - V) + I_{\mathrm{stim}}$$

$$\tau_\Theta\frac{\mathrm{d}\Theta}{\mathrm{d}t} = (\Theta_0 - \Theta)$$

Threshold: emit a spike if V crosses Θ
Reset: reset the membrane potential to V_r
increase Θ by b

Second variable: adaptive threshold

Practical implementation

...with some help from



Brian's approach

- *Philosophy*: Mathematical model descriptions
  - Flexible system to define models with equations
  - Takes care of numerical integration / synaptic propagation
  - Physical units
- *Technology*: Code generation
  - High-level descriptions transformed into low-level code
  - Transparent to user

More info

Website: https://briansimulator.org

Documentation: https://brian2.readthedocs.io

Discussion forum: https://brian.discourse.group

Articles:

Stimberg, Marcel, Romain Brette, and Dan FM Goodman. "Brian 2, an Intuitive and Efficient Neural Simulator." ELife 8 (2019): e47314. https://doi.org/10.7554/eLife.47314.

Stimberg, Marcel, Dan F. M. Goodman, Victor Benichoux, and Romain Brette. "Equation-Oriented Specification of Neural Models for Simulations." Frontiers in Neuroinformatics 8 (2014). https://doi.org/10.3389/fninf.2014.00006

Tutorial

Deepnote notebook

http://tiny.cc/2026-T09